1. Lab 09: IPv4, DHCP, and NAT

Lab Name: Lab 09: IPv4, DHCP, and NAT **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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AI Policy & Acceptable Use Guidelines


AI Policy & Acceptable Use Guidelines - Yellow Icon

Artificial Intelligence tools can be powerful learning aids, but they must be used ethically and responsibly to ensure you develop core networking skills.

- **Lab Assignments:** Do not use AI to perform lab steps or write your lab submissions. However, if you get stuck on a concept or technical obstacle, you may use AI as a virtual tutor to ask clarifying questions and help you troubleshoot.

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2. Background

Dynamic Host Configuration Protocol (DHCP) and Network Address Translation (NAT) are essential technologies for managing IPv4 addressing in modern networks. DHCP automates the assignment of IP addresses and network configuration parameters to devices, while NAT allows multiple devices on a private network to share a single public IP address, conserving the limited pool of globally routable IPv4 addresses. In this lab, you will configure both DHCP and NAT within your CML topology, observing how devices obtain IP addresses dynamically and how internal private addresses are translated for external communication.

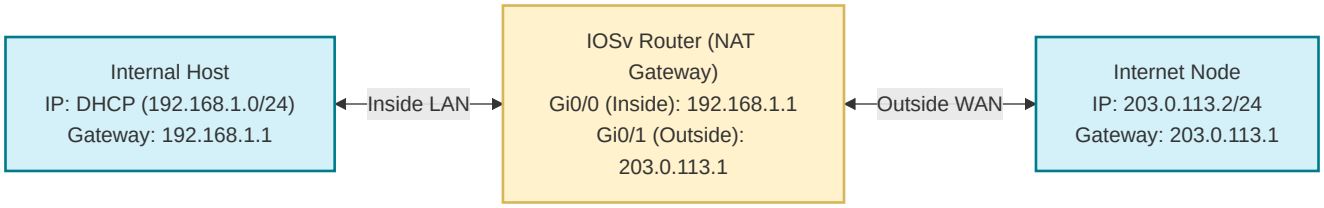
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3. Objective / Purpose

To configure a router as a DHCP server for local clients and implement Port Address Translation (NAT Overload) to allow private IPs to reach an external network.

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4. Topology Diagram



Details:

- 1x Router (NAT Gateway), 1x Internal Switch/Node, 1x External Node (representing the Internet).

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 1x IOSv Router, 2x Linux Nodes

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6. Configuration / Methodology

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	NAT Role
R1	G0/0 (Inside)	192.168.1.1	255.255.255.0	ip nat inside
R1	G0/1 (Outside)	203.0.113.1	255.255.255.0	ip nat outside

Key Configuration Snippets

```
! DHCP Config
ip dhcp pool LAN
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
! NAT Config
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

Narrative

We simulate an enterprise edge. The router dynamically assigns IPs to internal clients via DHCP. When clients attempt to ping the 'Internet' node, NAT overload translates their private source IP to the router's public outside interface IP.

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7. Verification & Testing

> *[!IMPORTANT]* ***REMINDER: All screenshots, console outputs, and packet captures (.pcap files) must be taken from the SAME active lab session. Mixing screenshots or trace files across different runs or sessions is not permitted.***

1. Output of `show ip dhcp binding` on R1.
2. Output of `show ip nat translations` after pinging the external node from the internal client.

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

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9. What to Hand In

- The Lab YAML file with your name, date, name of the lab at the top of the file in comments.
- The Wireshark packet capture file (`dhcp_nat_traffic.pcap` or `dhcp_nat_traffic.pcapng`) containing the captured DHCP negotiation and NAT translation traffic.
- A PDF with annotated screen shots showing the lab running, as well as screen shots of Wireshark captures of the lab showing DHCP negotiation and NAT translation traffic.